Data Analysis Project Report

PIZZA SALES ANALYSIS

Prepared By:

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🛠 Tools Used:

- SQL (MySQL)
- Excel (for visualization)
- Word (Documentation)

📌 1. Project Overview

This project focuses on analyzing pizza sales data to extract meaningful business insights. The dataset consists of multiple relational tables containing order details, pizza information, pricing, and categories.

The objective of this project is to:
Analyze sales performance
Identify customer preferences

Generate business insights using SQL queries

📊 2. Dataset Description

The dataset consists of 4 tables:

◈ Orders Table
Contains order ID, date, and time
Used for time-based analysis

◈ Order Details Table
Contains order_id, pizza_id, and quantity
Used to calculate total sales

◈ Pizzas Table
Contains pizza_id, size, and price
Used to calculate revenue

◈ Pizza Types Table
Contains pizza name, category, and ingredients
Used for category-level analysis

-- Basic:

-- 1. Retrieve the total number of orders placed.

```
SELECT COUNT(*) AS total_orders
FROM orders;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

total_orders
21350

-- 2. Calculate the total revenue generated from pizza sales.

```
SELECT ROUND(SUM(od.quantity * p.price), 2) AS total_revenue
FROM order_details od
JOIN pizzas p
ON od.pizza_id = p.pizza_id;s
select concat("$",round(sum(sales),2)) as Total_sale from sales_info ;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Total_sale
$817860.05

-- 3. Identify the highest-priced pizza.

```
SELECT pt.name, p.size, p.price FROM pizzas p
JOIN pizza_types pt
ON p.pizza_type_id = pt.pizza_type_id
ORDER BY p.price DESC
LIMIT 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

name	size	price
The Greek Pizza	XXL	35.95

-- 4. Identify the most common pizza size ordered.

```
SELECT p.size, COUNT(*) AS order_count FROM order_details od
JOIN pizzas p
ON od.pizza_id = p.pizza_id
GROUP BY p.size
ORDER BY order_count DESC
LIMIT 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

size	order_count
L	18526

-- 5. List the top 5 most ordered pizza types along with their quantities.

```
SELECT pt.name, SUM(od.quantity) AS total_quantity FROM order_details od
JOIN pizzas p ON od.pizza_id = p.pizza_id
JOIN pizza_types pt ON p.pizza_type_id = pt.pizza_type_id
GROUP BY pt.name
ORDER BY total_quantity DESC
LIMIT 5;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

name	total_quantity
The Classic Deluxe Pizza	2453
The Barbecue Chicken Pizza	2432
The Hawaiian Pizza	2422
The Pepperoni Pizza	2418
The Thai Chicken Pizza	2371

-- Intermediate:

-- 6. Join the necessary tables to find the total quantity of each pizza category ordered.

```
SELECT pt.category, SUM(od.quantity) AS total_quantity FROM order_details od
JOIN pizzas p
```

```
ON od.pizza_id = p.pizza_id
JOIN pizza_types pt
ON p.pizza_type_id = pt.pizza_type_id
GROUP BY pt.category
ORDER BY total_quantity DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

category	total_quantity
Classic	14888
Supreme	11987
Veggie	11649
Chicken	11050

-- 7. Determine the distribution of orders by hour of the day.

```
SELECT EXTRACT(HOUR FROM time) AS hour,
    COUNT(order_id) AS total_orders
FROM orders
GROUP BY hour
ORDER BY hour;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

hour	total_orders
9	1
10	8
11	1231
12	2520
13	2455
14	1472
15	1468
16	1920
17	2336
18	2399
19	2009
20	1642
21	1198
22	663
23	28

-- 8. Join relevant tables to find the category-wise distribution of pizzas.

```
SELECT pt.category, COUNT(*) AS total_orders
FROM order_details od
```

```
JOIN pizzas p
ON od.pizza_id = p.pizza_id
JOIN pizza_types pt
ON p.pizza_type_id = pt.pizza_type_id
GROUP BY pt.category;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

category	total_orders
Classic	14579
Veggie	11449
Supreme	11777
Chicken	10815

-- 9. Group the orders by date and calculate the average number of pizzas ordered per day.

```
SELECT ROUND(AVG(total_pizzas), 2) AS avg_pizzas_per_day
FROM (
    SELECT o.date, SUM(od.quantity) AS total_pizzas
    FROM orders o
    JOIN order_details od
    ON o.order_id = od.order_id
    GROUP BY o.date
) daily_orders;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

avg_pizzas_per_day
138.47

-- 10. Determine the top 3 most ordered pizza types based on revenue.

```
SELECT pt.name,
       SUM(od.quantity * p.price) AS revenue
FROM order_details od
JOIN pizzas p
ON od.pizza_id = p.pizza_id
JOIN pizza_types pt
ON p.pizza_type_id = pt.pizza_type_id
GROUP BY pt.name
ORDER BY revenue DESC
LIMIT 3;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

name	revenue
The Thai Chicken Pizza	43434.25
The Barbecue Chicken Pizza	42768
The California Chicken Pizza	41409.5

-- Advanced:

-- 11. Calculate the percentage contribution of each pizza type to total revenue.

```
SELECT pt.name,
 ROUND(SUM(od.quantity * p.price) * 100 /
 (SELECT SUM(od2.quantity * p2.price)
  FROM order_details od2
   JOIN pizzas p2
  ON od2.pizza_id = p2.pizza_id), 2) AS percentage FROM order_details od
JOIN pizzas p
ON od.pizza_id = p.pizza_id
JOIN pizza_types pt
ON p.pizza_type_id = pt.pizza_type_id
GROUP BY pt.name
```

ORDER BY percentage DESC;

Result Grid | Filter Rows:

name	percentage
The Thai Chicken Pizza	5.31
The Barbecue Chicken Pizza	5.23
The California Chicken Pizza	5.06
The Classic Deluxe Pizza	4.67
The Spicy Italian Pizza	4.26
The Southwest Chicken Pizza	4.24
The Italian Supreme Pizza	4.09
The Hawaiian Pizza	3.95
The Four Cheese Pizza	3.95
The Sicilian Pizza	3.78
The Pepperoni Pizza	3.69
The Greek Pizza	3.48
The Mexicana Pizza	3.27
The Five Cheese Pizza	3.19
The Pepper Salami Pizza	3.12

Result Grid | Filter Rows:

name	percentage
The Greek Pizza	3.48
The Mexicana Pizza	3.27
The Five Cheese Pizza	3.19
The Pepper Salami Pizza	3.12
The Italian Capocollo Pizza	3.07
The Vegetables + Vegetable...	2.98
The Prosciutto and Arugula ...	2.96
The Napolitana Pizza	2.95
The Spinach and Feta Pizza	2.85
The Big Meat Pizza	2.81
The Pepperoni, Mushroom, ...	2.3
The Chicken Alfredo Pizza	2.07
The Chicken Pesto Pizza	2.04
The Soppressata Pizza	2.01
The Italian Vegetables Pizza	1.96

Result 1

-- 12. Analyze the cumulative revenue generated over time.

```
SELECT o.date,
    SUM(od.quantity * p.price) AS daily_revenue,
    SUM(SUM(od.quantity * p.price))
    OVER (ORDER BY o.date) AS cumulative_revenue
FROM orders o
JOIN order_details od
ON o.order_id = od.order_id
JOIN pizzas p
ON od.pizza_id = p.pizza_id
GROUP BY o.date
ORDER BY o.date;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

date	daily_revenue	cumulative_revenue
2015-01-01	2713.8500000000004	2713.8500000000004
2015-01-02	2731.8999999999996	5445.75
2015-01-03	2662.4	8108.15
2015-01-04	1755.4500000000003	9863.6
2015-01-05	2065.95	11929.55
2015-01-06	2428.95	14358.5
2015-01-07	2202.2000000000003	16560.7
2015-01-08	2838.3499999999995	19399.05

Result 2

-- 13. Determine the top 3 most ordered pizza types based on revenue for each pizza category.

```
SELECT category, name, revenue FROM (
 SELECT pt.category, pt.name, SUM(od.quantity * p.price) AS revenue,
 RANK() OVER (
  PARTITION BY pt.category
            ORDER BY SUM(od.quantity * p.price) DESC ) AS rnk
   FROM order_details od
   JOIN pizzas p ON od.pizza_id = p.pizza_id
   JOIN pizza_types pt   ON p.pizza_type_id = pt.pizza_type_id
   GROUP BY pt.category, pt.name
) ranked
WHERE rnk <= 3;
```

category	name	revenue
Chicken	The Thai Chicken Pizza	43434.25
Chicken	The Barbecue Chicken Pizza	42768
Chicken	The California Chicken Pizza	41409.5
Classic	The Classic Deluxe Pizza	38180.5
Classic	The Hawaiian Pizza	32273.25
Classic	The Pepperoni Pizza	30161.75
Supreme	The Spicy Italian Pizza	34831.25
Supreme	The Italian Supreme Pizza	33476.75
Supreme	The Sicilian Pizza	30940.5
Veggie	The Four Cheese Pizza	32265.70000000065
Veggie	The Mexicana Pizza	26780.75